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Energy Saving Algorithm of HVAC System Based on Deep Reinforcement Learning with Modelica Model

Yuzhe Zhang, Qianchuan Zhao*

Center for Intelligent and Networked Systems,

Department of Automation, BNRist, Tsinghua University, Beijing 100084, China
E-mail: zhaoqc@tsinghua.edu.cn

Abstract: Aiming at the energy-saving control problem of Heating, Ventilation and Air Conditioning (HVAC) system, a Deep Reinforcement Learning (DRL) control method based on Deep Deterministic Policy Gradient (DDPG) is proposed. The optimization capability of DDPG is improved by collecting sensor data inside the HVAC system and using Autoencoder (AE) for feature extraction. DDPG improves the system Coefficient of Performance (system COP) by controlling the mass flow of the chilled water pump of the HVAC system. This method is simulated, trained and verified on the refined HVAC system model built in this paper. The experimental results show that the method proposed can effectively improve the system COP and achieve the goal of saving energy in the HVAC system.

Key Words: HVAC, Modelica, DDPG, Autoencoder

1 Introduction

In the face of severe climate change, tremendous energy shortage and increasing economic demand, the rapid growth of building energy consumption has become the main challenge of current building energy efficiency. From a global perspective, the contribution from buildings towards total energy consumption has reached about 30% [1]. The energy use of the HVAC system accounts for about 40% of the entire building energy consumption [2]. Therefore, it is significant to study the energy-saving control of HVAC system.

Usually, typical HVAC systems mainly contain cooling towers, chillers, fan coil units and pumps [3]. Cooling towers reduce the temperature of cooling water by exchanging heat with the air. The chilled water flowing out of the chiller is pressurized by the chilled water pump and sent into the fan coil unit to exchange heat with the room. A typical HVAC system structure is shown in Figure 1.

At present, the research on HVAC system energy consumption focuses on control solutions based on physical models and driven by data [4]. Reinforcement Learning (RL), as a field of machine learning dedicated to control problems, is becoming a promising HVAC system control method. [5] applied Q-Learning to control pumps and cooling tower fans of the HVAC system, improving the COP of the system. In [6], DQN was utilized to control the temperature set point of HVAC systems in a building area to achieve energy-cost savings with the presence of changing electricity prices. [7] generated an optimal DDPG-based control method for residential HVAC systems, minimizing energy consumption. Through the HVAC model built by EnergyPlus, [8] applied the Soft Actor Critic control method to adjust the temperature set point of the HVAC system. [9] designed the HVAC system controller by combining Long-Short-Term Memory (LSTM) algorithm with Actor-Critic, which improved energy efficiency by an average of 2.5% compared to traditional control strategies. Based on the HVAC model built by TRNSYS, [10] proposed a control strategy based on DDPG, which reduced the energy consumption of the HVAC system by 4.31%.

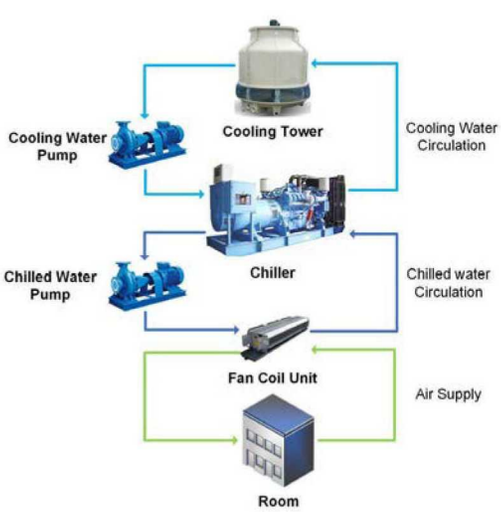


Fig. 1: Typical HVAC system structure chart

The HVAC system is highly complex, and many types of sensor data can be collected in the actual system. However, in current researches on applying RL to the field of HVAC system control, the state space is mainly constructed by external data of the HVAC system, such as room temperature, weather temperature and electricity price. Consequently, previous studies lack the collection and processing of internal sensor data inside the HVAC system, such as cooling water temperature, the temperature and flow

* corresponding author: Qianchuan Zhao

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of chilled water. One reason for the absence is that there are many types of sensors in the HVAC system, and the data varies greatly. Moreover, methods of processing these sensor data are lacking. Another reason is the deficiency of a sufficiently refined model of the HVAC system, which makes RL difficult to obtain enough sensor data and control the parameters of equipment. Consequently, it is quite vital to design a reinforcement learning control algorithm of HVAC system on the basis of sensor data in larger quantity and varieties.

As for contributions in this paper, above all, the algorithm on the basis of DDPG for energy-saving control of HVAC system is proposed. Combining a feature extraction method of HVAC system based on AE, the DDPG algorithm enhances its optimization effect. Furthermore, a calibrated refined HVAC system model is employed for training and experimentation of the energy-saving algorithm mentioned above, which enables processing multiple types of sensor data in HVAC system.

2 Deep Reinforcement Learning Control

The Figure 2 shows the framework of AE and DDPG proposed in this paper. When the control policy is executed, the AE encodes the collected sensor data of the HVAC system, then inputs it to the DDPG, and finally the DDPG outputs the action to adjust the mass flow of the chilled water pump of the HVAC system.

In this section, firstly, the RL environment including the reward function, state space and action space is designed. Next, the AE is introduced to encode the features of the HVAC system. Finally, the DDPG is designed to control the chilled water pump of the HVAC system.

2.1 Reinforcement Learning Model

RL pays attention to how an agent (e.g. an HVAC system controller) can take action (e.g. adjust the temperature set point) when it is in a state (e.g. the current room temperature is 26°C) in an environment (e.g. an HVAC system) to maximize/minimize some incentives (e.g. occupant's thermal comfort). The goal of RL is to learn the optimal action an agent can take in each state. DRL, as a branch of RL, approximates the policy function by using deep neural networks to handle continuous states and actions.

- Rewards

The optimization goal of energy-saving control of HVAC system is to improve the system COP. Thus, the reward function includes the system COP [11]. Considering the energy-consuming equipment of the HVAC system model mainly includes cooling towers, pumps and water chillers, the system COP is set as the quotient of the cooling capacity of the chilled water circulation and the sum of the powers of the cooling towers, pumps and water chillers. Under the control of the RL, the temperature of terminal room often deviates from the expected value, which affects the cooling effect of the air conditioner. Therefore, a temperature offset penalty term needs to be added to reward function so that a large deviant of the room temperature could be avoided.

In summary, the reward function of RL is designed as the sum of the System COP and a temperature offset penalty term, as shown in formula (1). The purpose of the reward function is to minimize energy consumption while maintaining the room temperature within a pre-set range. In the final analysis of the optimization effect of different algorithms, the system COP of the HVAC system will be analyzed separately.

$$Reward = \frac{Q_{CHW}}{P_{chillers} + P_{towers} + P_{pumps}} + \lambda \left| T_{room} - T_b \right| \quad (1)$$

Q_{CHW} is the cooling capacity of the chilled water circulation, P_{towers} is the power of the cooling tower, $P_{chillers}$ is the power of the water chillers, and P_{pumps} is the sum of powers adding by the cooling water pump and the chilled water pump. The second term of the formula is the temperature offset penalty, and λ is a negative constant ($-1 < \lambda < 0$) to prevent terminal room temperature from deviating too much from the desired temperature.

- State Space

In the HVAC system model, the external sensor data of the HVAC system mainly includes weather data and room temperature, and the internal sensor data majorly contains equipment operation data and pipeline sensor data. Because the weather temperature will directly affect the cooling conditions of the cooling tower, the external weather data is added to the state space. Combined with the sensor types and deployment locations in the actual system, the set state space consists of following seven parts: T_{wea} (weather temperature), T_{cw} (leaving temperature of cooling water),

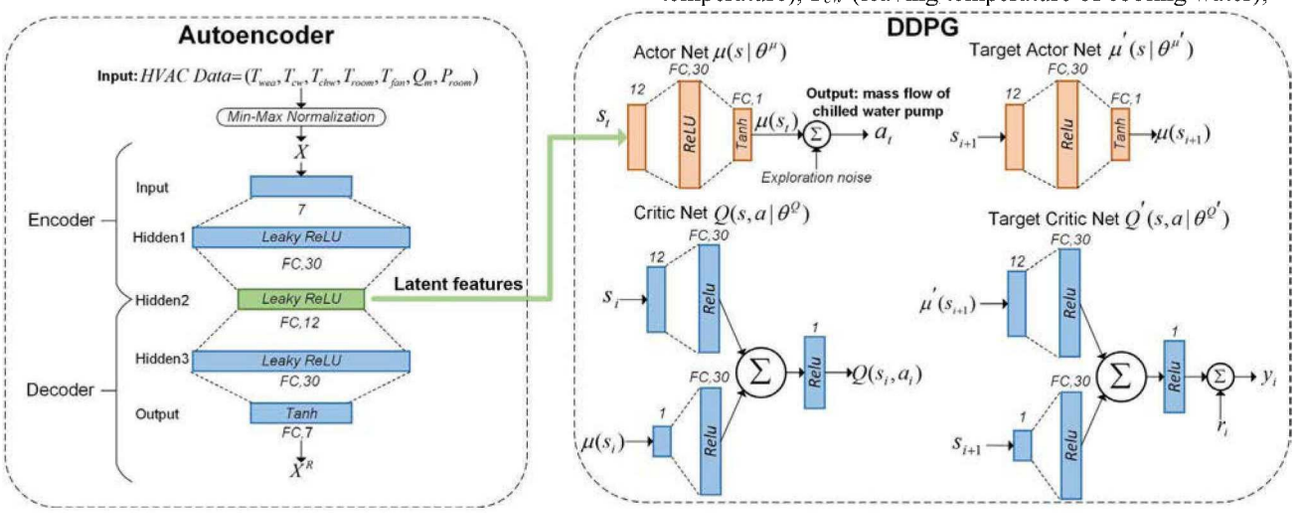


Fig. 2: Framework of AE and DDPG

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T_{chw} (return temperature of chilled water), T_{fan} (air supply temperature of terminal room), T_{room} (terminal room temperature), Q_m (mass flow of chilled water pump) and P_{room} (heat production power of terminal room), as shown as follow:

$$State=(T_{wea},T_{cw},T_{chw},T_{room},T_{fan},Q_m,P_{room}) \quad (2)$$

- Action Space

In the actual HVAC system, the set values of water chillers and cooling towers usually do not change frequently, while the set value of pumps are always controlled. Therefore, the action of RL is designed as the change value of the mass flow of the chilled water pump, such as $\pm$ 0.1kg/s.

Each device in the actual HVAC system is subject to safety constraints. If the set point of the device exceeds the safety boundary, it will not only cause the room temperature to fail to meet the requirements, but also lead to system instability and even equipment damage, resulting in losses of equipment and property. In the exploration and learning process of RL, the action output value will cause the set value of the mass flow of the chilled water pump to deviate from the safety requirements. If no safety constraints are imposed on the action output value of RL, the algorithm effect will be deteriorated, and the simulation results will also not match the actual system. Agents are expected to improve the system COP within a limited action space. This study will combine the safety constraints of the HVAC system with the training process of the DRL: the action space, that is, the mass flow of the chilled water pump, is limited within the range of 80% to 120% of the reference value. For example, the mass flow reference value of the chilled water pump is 2kg/s, and then the mass flow range of the chilled water pump, that is, the safety margin, is set to be 1.6kg/s~2.4kg/s. When the action output value deviates from the safe boundary of the chilled water pump, on one hand, in order to prevent this from happening again, the reward of the current state will be set as a negative constant. On the other hand, current Eposide training will be brought to an end, the mass flow of the chilled water pump will be reset to the reference value, and another brand new Eposide training will arrive in succession. The safety boundary processing flow is shown in Figure 3.

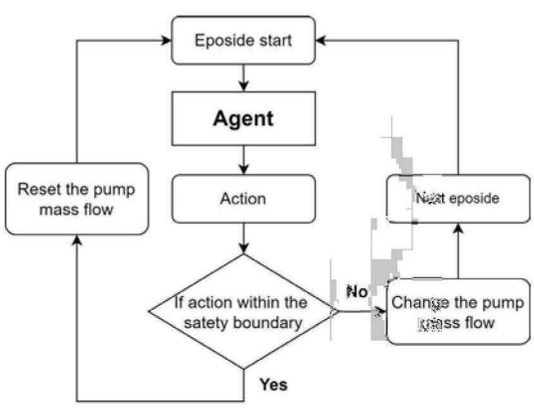


Fig. 3: Flow chart of safety boundary processing

2.2 Features Extraction via Autoencoder

In deep learning, AE is an unsupervised neural network model [12]. ,AE is widely used in feature learning of sample data in DL, which can learn the latent features of the input data. Meanwhile, AE could apply the latent features to reconstruct the original input data. By compressing the original input feature x, decompressing it into r, and then comparing x and r to calculate the prediction error, and finally performing backpropagation, the accuracy of auto-encoding is gradually improved. The middle part of the trained AE is able to summarize the essence of the original data. Thus, the encoder is usually encapsulated as a model for calling.

AE can act as a feature extractor. The processing method of inputting static volume of the HVAC system directly into the RL model as a feature for training and calculation has two shortcomings. One is that the input data does not reflect the correlation between the features. Another shortcoming is that input features are not preprocessed. Different feature dimensions will bring about great difference in the magnitude of feature values, which in turn affects the optimization effect of RL model. Therefore, AE is adopted in this paper to solve the above two deficiencies. Above all, the original state space data of the HVAC system is input into the AE for training. Then, the output of the AE hidden layer of the AE is taken as the new state space, and finally it is input into the DDPG for training and calculation.

In this paper, the number of network layers of the AE is set to 5 layers. The input layer and the output layer each have 7 neurons, which is the same as the state space dimension. After the original state space data of HVAC system is encoded by AE, the hidden layer of AE outputs a 12-dimensional vector as the input of DDPG.

2.3 Deep Deterministic Policy Gradient(DDPG)

DRL can solve the problem of large state space and action space by combining deep learning and reinforcement learning theory, which overcomes the shortcomings of traditional RL method [13]. DRL uses neural networks to represent value functions.

DDPG is a deep reinforcement learning algorithm that combines deep neural networks and policy gradient [14]. DDPG adopts Actor-Critic architecture. Actor and Critic are composed of trained network and target network respectively, which is constantly equivalent to a total of 4 deep neural networks, namely Actor Net, Cirtic Net, Actor Target Net and Cirtic Target Net.

The actor net is updated by t sampled graient:

$$\nabla_{\theta^\mu}J\approx\frac{1}{N}\sum_i\nabla_a Q(s,a|\theta^Q)|_{s=s_i,a=\mu(s_i)}\nabla_{\theta^\mu}\mu(s|\theta^\mu)|_{s_i} \quad (3)$$

The critic net is updated by minimizing this loss function:

$$L=\frac{1}{N}\sum_i(y_i-Q(s_i,a_i|\theta^Q))^2 \quad (4)$$

DDPG uses the Experience Buffer mechanism to store the data that the Agent interacts with the environment. Different from the delayed replication of existing networks between DQN target networks, DDPG uses soft update to iterate the parameters in the two target networks and adds a gaussian noise to the actions taken. In general, DDPG can handle practical problems with high stability and precision.

Considering DDPG can output continuous values, it is suitable as an algorithm for controlling chilled water pumps. As a result, the Actor network of DDPG is designed as a

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2-layer fully-connected neural network. The Critic network design is set as a 3-layer fully-connected neural network. Plus, Actor Net and Actor Target Net, Critic Net and Critic Target Net have the same network structure. The detailed training principles of DDGP and AE are shown as followings in Figure 4.

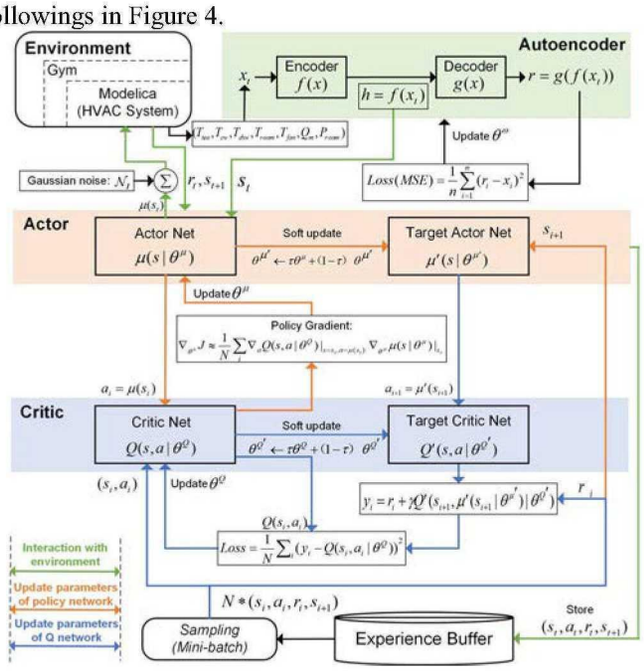


Fig. 4: Flow chart of DDPG training

The lantent features, extracted by AE from the state space of the HVAC system model, function as the input of the DDPG model which will accordingly output the mass flow set value of the chilled water pump. During the training process, collected from the HVAC system model, these long-term simulation data will be employed to implement the pre-training of AE and DDPG as the basis. In this paper, an online training is executed by AE and DDPG.

3 Modeling and Experiments

3.1 Modeling of HVAC System

The establishment of the HVAC system model can provide an experimental environment and a verification platform for the control algorithm. Modelica supports co-simulation with Python and the improvement of the HVAC model library. In this paper, on the basis of Modelica Standard Library and Modelica Bulidings Library [15], OpenModelica is used to model a HVAC system. The final HVAC system model has a cooling tower, a cooling water pump, a chiller, a chilled water pump, a fan coil unit, a fan, a room and several controllers.

In the Modelica model, the right side is the HVAC system and the left side is the controller. As shown in Figure 5, the right HVAC system section includes equipment and sensors, and the left controller section includes controllers and data transmission routes. Among them, the blue route transmits external data of the HVAC system (e.g. weather data), and the green route transmits internal sensor data of the HVAC system (e.g. chilled water leaving temperature). Values of the chilled water pump controller and the room heat production power controller are output by the Python program.

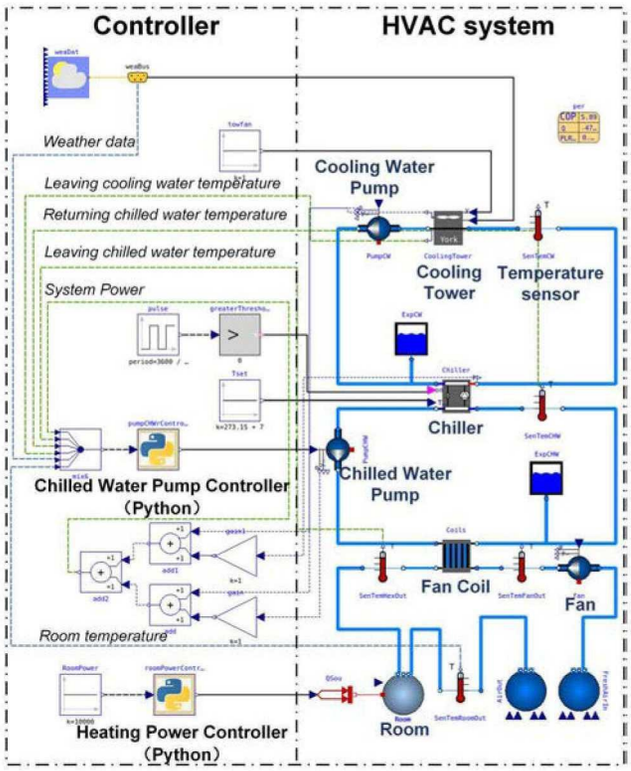


Fig. 5: HAVC system model based on Modelica

This paper collects the equipment parameters and sensor data from a set of HVAC systems deployed in actual buildings in Sichuan Province, and then combines them as a data set to be the solid data foundation for the calibration and simulation verification of our model. The error of model can be evaluated by calculating the Mean Bias Error (MBE) per hour and the Coefficient of Variation of the root Mean Square Error (CV (RMSE)) . The calculation formulas of MBE and CV (RMSE) are shown in formula (6) and formula (7) respectively. Among them, is the simulated value, is the actual value, and is the average value. The Principle 14 of ASHRAE [16] states that when the hourly MBE is within ±10% and the CV (RMSE) is within ±30%, the model accuracy can be considered to meet the calibration criteria.

$$MBE = \frac{\sum(\hat{Y}_i - Y_i)}{\sum \hat{Y}_i} \tag{5}$$

$$CV(RMSE) = \frac{\sqrt{\sum(\hat{Y}_i - Y_i)^2 / n}}{\bar{Y}} \tag{6}$$

In this study, the equipment parameters of the Modelica model are calibrated with the parameters in the HVAC system dataset. The cooling water leaving temperature in the dataset is input into the HVAC system model to analyze whether the chilled water return temperature in the simulation results is consistent with the data source. In the comparison, the MBE of the chilled water return temperature is 3.61%, and the CV (RMSE) is 3.05%. The simulation results are shown in Figure 6. Experimental results verify that the accuracy of the HVAC system model meets ASHRAE Principle 14.

Developed by OpenAI, Gym is a platform for researching, developing and comparing RL-related algorithms, providing a standard RL training environment to train and

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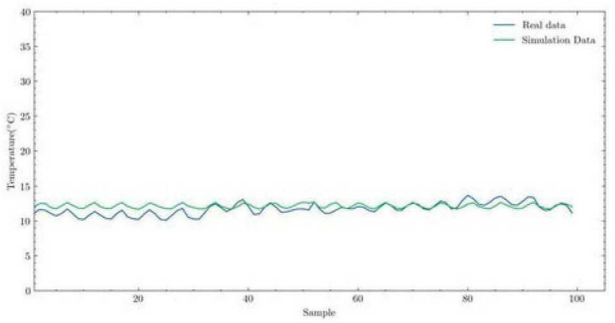


Fig. 6: Comparison of simulation results of the cooling water return temperature

test various RL algorithms [17]. Therefore, secondary development is carried out based on the Modelica model, and Python is used to encapsulate the Modelica model as a Gym environment.

3.2 Experimental Setting

Experiments require setting a baseline environment. The room heat production power is set to 10KW. In order to ensure that the room temperature is within the required range, the main parameters of each equipment in the HVAC system are set to the data shown in Table 1, and these parameters are fixed as the baseline of the experiment. This paper fixes the parameters of all equipment except the chilled water pump and uses the continuous value output by RL as the mass flow value of the chilled water pump.

The weather data used in this model is actual built-in weather data in the Modelica Building Library.

Table 1: Baseline values of main parameters of HVAC system

Main parameters of equipment	Baseline values
Cooling Tower Approach Temperature	4°C
Chilled water leaving temperature	7°C
Mass flow of cooling water pump	4kg/s
Mass flow of chilled water pump	2kg/s
Mass flow of fan air	2kg/s
Room heat production power	10KW

3.3 Analysis of Simulation Results

First, the feature extraction effectiveness of AE is verified. The heat production power of the room is set to 5KW, 10KW and 15KW. The HVAC system data under the three working conditions were collected and input into the AE for training respectively, and it was observed whether the hidden layer output of the auto-encoder could successfully classify the HVAC system state space of the three working conditions. The number of hidden layer neurons of the AE is set to 3, the training time is set to 2000 hours, and the sampling period is set to 15 minutes. The static volume of the HVAC system under the three working conditions are taken separately and then input into the AE for training. After AE completes training, 200 HVAC system sampling point data are input into AE, and the output vector of the hidden layer is visualized. The results are shown in Figure 7. It can be seen that AE can effectively map the sensor data of the HVAC system under the three working conditions to different regions, which verifies the feature extraction ability of AE.

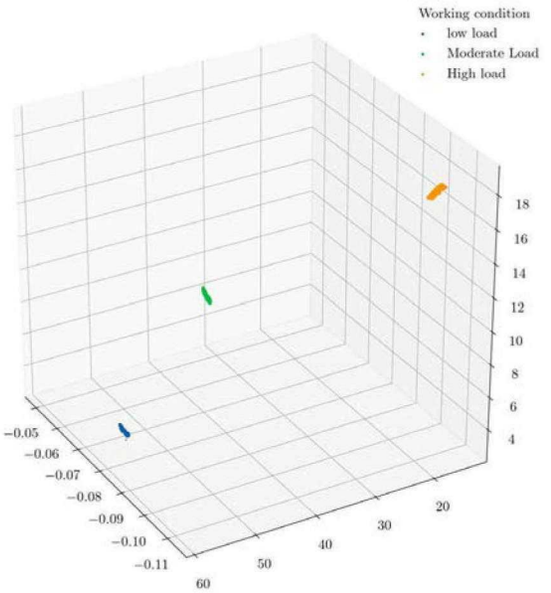


Fig. 7: Visualized 3-dimensional vector generated by AE hidden layer

Next, the influence of sensor data on DDPG is verified. The optimization effect of DDPG is compared in two cases where the state space is only composed of external sensor data (2-dimensional vector) or the state space combines the external sensor data with the internal sensor ones (7-dimensional vector). The DDPG algorithm is implemented in Python code, and the Modelica training time is set to 2000 hours. The time step is set to 15 minutes, and the co-simulation of Python and Modelica is executed. The experimental results are shown in Figure 8 and Figure 9. When the state space is composed of only 2-dimensional vectors, the System COP is improved by 3.43% compared to the baseline. When the state space is composed of 7-dimensional vectors, the System COP is improved by 5.31% compared with the baseline, which verifies that the introduction of internal sensor data to construct the state space improves the optimization effect of DDPG.

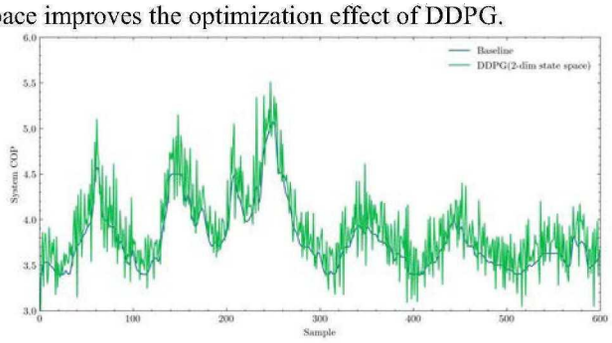


Fig. 8: DDPG optimal result of the state space composed of 2-dimensional vector

Finally, the effect of feature extraction by AE on the optimization effect of DDPG is verified. Taking the hidden layer output of AE as the input of DDPG, the experimental results are shown in Figure 10. Compared with 5.36% improvement of only using DDPG, the System COP is improved by 7.41% over the baseline when it comes to the

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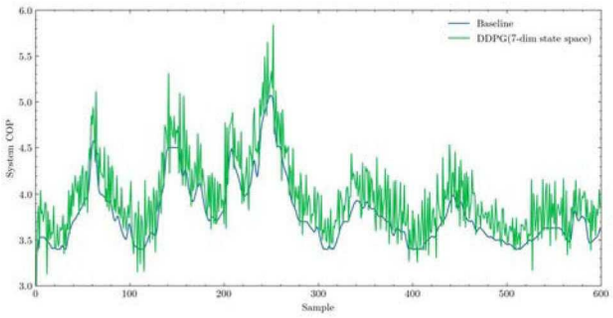


Fig. 9: DDPG optimal result of the state space composed of 7-dimensional vector

combination of AE and DDPG, which is a significant amelioration. This verifies the feature extraction ability of AE, and reflects the advantages of the AE to secondarily construct the state space. The comparison of the optimization results of all control methods is shown in Figure 11.

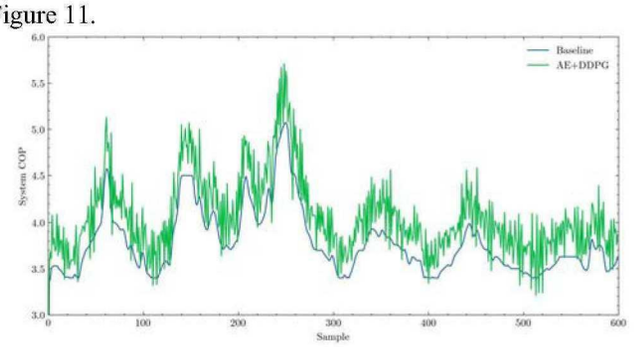


Fig. 10: Optimal result of the 'AE+DDPG'

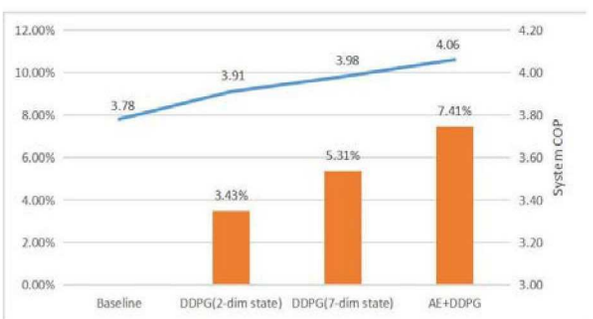


Fig. 11: Comparison of the system COP of baseline, DDPG and 'AE+DDPG'

4 Conclusions

In the field of HVAC system energy-saving optimization, this paper finds that the HVAC system control method based on AE and DDPG improves the System COP by adjusting the mass flow of the chilled water pump. The proposed method of constructing the state space based on the internal sensor data of the HVAC system and the method of introducing AE for feature extraction both improve the optimization effect of DDPG. The refined HVAC system model built on Modelica supports the realization and verification of the DRL control algorithm. To sum up, the control method proposed in this paper can achieve the goal of saving energy in the HVAC system.

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